

Homework (Habanero)

- Q1.** A doctor prescribes 15 pills to a patient, with 3 pills to be taken daily. How many days will the pills last?
(A) 3 days
(B) 4 days
(C) 5 days
(D) 6 days
(E) 7 days
- Q2.** A biologist is studying cell growth. If a cell divides into 2 daughter cells every 5 hours, how many cells will there be after 15 hours?
(A) 4 cells
(B) 6 cells
(C) 8 cells
(D) 12 cells
(E) 16 cells
- Q3.** What is the remainder when 27 is divided by 9?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 6
- Q4.** A medical trial has 120 participants. If they are divided into groups of 8, how many groups can be formed?
(A) 12 groups
(B) 15 groups
(C) 16 groups
(D) 20 groups
(E) 24 groups
- Q5.** A pharmaceutical company produces 210 units of a medication daily. If 5 units are packaged in each bottle, how many bottles are filled daily?
(A) 40 bottles
(B) 41 bottles
(C) 42 bottles
(D) 45 bottles
(E) 50 bottles
- Q6.** What is the quotient and remainder when 48 is divided by 7?
(A) 6 R 6
(B) 6 R 5
(C) 7 R 1
(D) 8 R 2
(E) 9 R 3
- Q7.** A researcher is collecting data on the average lifespan of a species. If the data shows that the lifespan is 25 years, and this value is 5 more than a multiple of 4, what is the remainder when the multiple is divided by 9?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q8.** A hospital has 180 beds. If 15 beds are allocated to each ward, how many wards can be created?
(A) 10 wards
(B) 11 wards
(C) 12 wards
(D) 13 wards
(E) 14 wards

Open Ended Questions (FRQ)

- Q9.** A medication requires $12 \cdot 5 + 3$ milligrams of an active ingredient per dose. If 3000 milligrams are available, determine how many doses can be prepared, and show the remainder.
- Q10.** In a study on population growth, it is observed that the number of cells doubles every 6 hours. If there are initially 24 cells, calculate the number of cells after 24 hours, expressing the answer in terms of the number of cells and the time elapsed.

Chef's Tips

- Tip 1: Provide clear instructions for calculations and graphing
- Tip 2: Encourage students to use examples for verification
- Tip 3: Discuss divisibility rules to apply to word problems